

V A N E

Zips jam. *Feathers don't.*

A closure with no slider, no teeth and no metal, taken from the way a bird's feather locks itself back together. Self-initiated by made. by ac, built whole, and honest about being a concept.



One tape. No slider. *One material.*

A zip is a slider, two tapes and a few hundred teeth that all have to stay in register. VANE is a single extruded polymer tape carrying a row of hooked barbs along its edge. Two tapes face each other. That is the entire bill of materials.

01 · What it does

Closes anywhere, opens with one hand

There is no box and pin, so there is no place you have to start. Press the two edges together at any point along the seam, or run a finger down it. To open, lift a corner and peel. Either end, either direction, gloves on, in the dark.

02 · Why it holds

Strong where a garment pulls, weak where a hand pulls

Load along the seam drives each hook further into its ridge, so the closure gets stronger exactly when it is loaded. Peel is the one direction the geometry gives up, and it releases in sequence rather than all at once.

42 N

SHEAR HOLD / 100 MM

3.1 N

PEEL RELEASE

14×

HOLD TO RELEASE RATIO

0

METAL PARTS

The last serious idea about closing things was *a hundred and thirteen* years ago.

Gideon Sundback fixed Whitcomb Judson's clasp locker in 1913 and the object has barely moved since. We put it on every bag, every coat, every tent and every pair of trousers, and we quietly organise our lives around its three failures.

01

It jams

Hundreds of parts that must stay in register. Any one can fail, and when one does the whole garment is finished. Nothing else on a jacket fails this often.

02

It excludes

Two hands, a pinch grip, and enough fine motor control to start the box and pin. Arthritis takes that away. So does a tremor, a prosthesis, an infant, or cold hands.

03

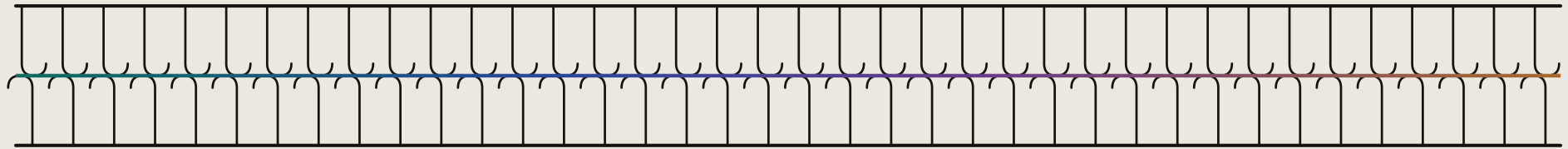
It outlives the garment

A metal zip on a cotton coat makes that coat mixed waste. Recyclers cut every closure out by hand, so most of them do not bother.

Nature shipped this *a long time ago*.

Velcro came from a burr stuck to a dog. Nobody went back for the better idea on the same walk, which was the bird overhead.

HOOKED BARBULES



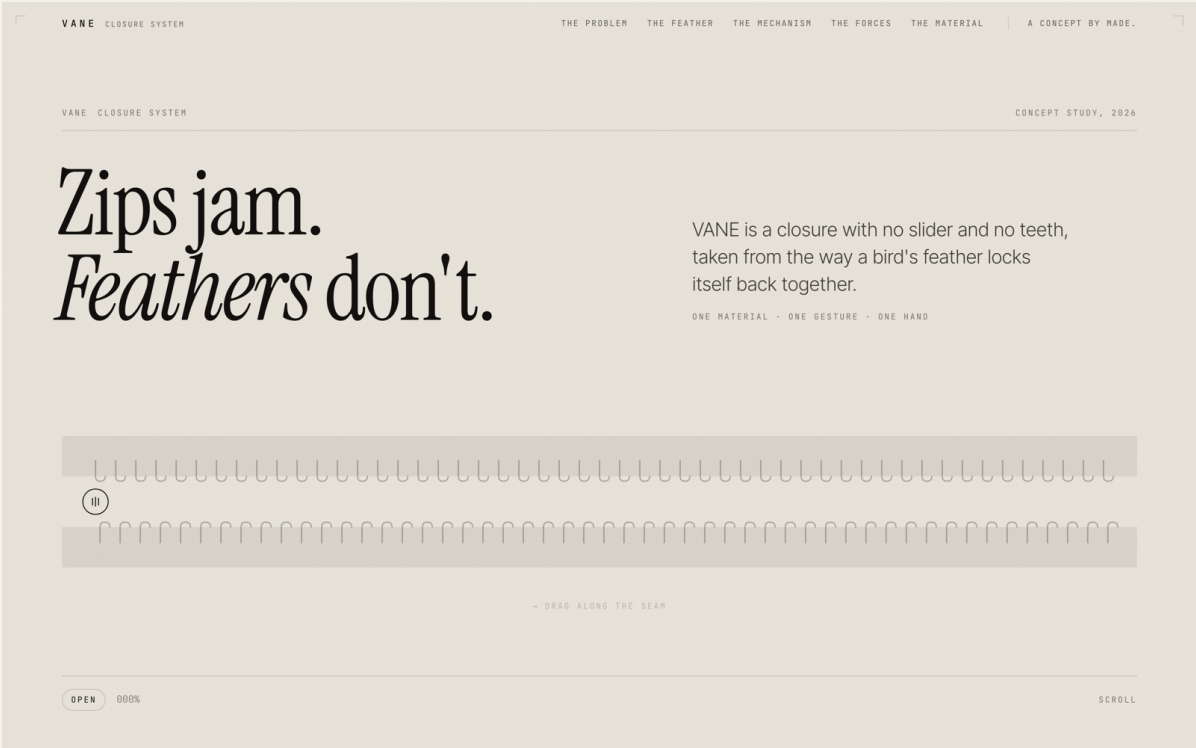
RIDGED BARBULES

A feather vane holds against a gale, opens without damage, and re-links in a single pass of the beak. The structure reassembles itself because the geometry only fits one way. That is the property we wanted: a closure nobody has to align.

The seam *is* the interface.

The hero is not a picture of the product, it is the product. Drag along the seam and it closes as a wavefront rather than a snap, with the tape edge rippling behind the front, because that is what a feather does when it goes through a beak.

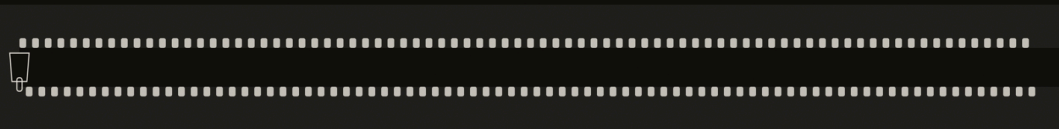
Iridescence fills in behind the wavefront and nowhere else, so the colour is a readout of how much structure has locked.



HERO · DRAG-TO-SEAL SEAM · SVG AND REQUESTANIMATIONFRAME

The last serious idea about closing things was *a hundred and thirteen* years ago.

Gideon Sundback fixed Whitcomb Judson's clasp locker in 1913 and the object has barely moved since. We put it on every bag, every coat, every tent and every pair of trousers, and we quietly organise our lives around its failures. Here is one.



TRAVELLING | DRAG THE SLIDER

THE INCUMBENT · A WORKING ZIP THAT REFUSES AT 62 PERCENT

A zip that jams, *on purpose*.

We built a working zip into the argument. It travels, it catches at sixty two percent, it throws its teeth out of register, and on the second attempt the label tells you it will do this again.

Frustrating the visitor is the point. Everyone already knows this object fails. Making them feel it, in the first thirty seconds, does more than any statistic we could have put on the page.

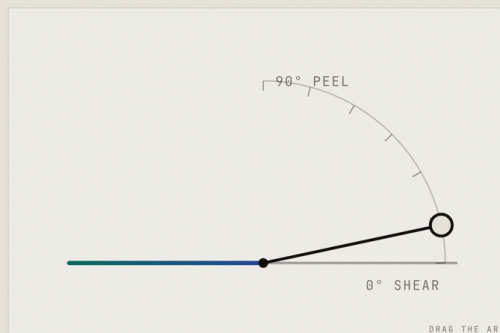
The physics, made draggable.

Pull the arm from zero to ninety degrees and watch hold force collapse along a live curve.

A coat pulls on its closure at nought to about fifteen degrees, all day. A hand lifting one corner pulls at ninety. The seam holds fourteen times harder against the first than the second, and neither number is a compromise. That gap is the entire product.

Strong one way. *Weak* the other. On purpose.

A closure does not need to be strong in every direction. It needs to be strong in the directions a garment pulls, and weak in the one direction a hand pulls. Those are not the same direction, and that gap is the entire product.



HOLD FORCE

39.8 N

12° SHEAR

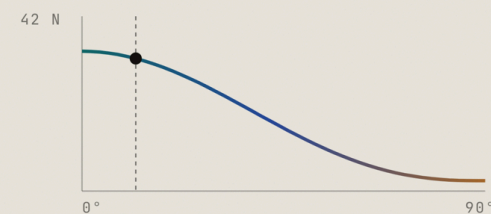


PLATE IV ON THE SITE • LIVE FORCE RIG, 100 MM TEST SEAM

A natural history plate, crossed with a *materials lab*.

BONE · ECE8DF	INK · 14130F	TEAL · 0D6D62	BLUE · 27499B	VIOLET · 6B3A8E	BRONZE · A96A2C
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The one rule the whole system hangs on: structural iridescence is the only colour, and it appears only where the barbs have locked. A feather's colour comes from structure rather than pigment, so the seam has no colour until it has order. Everywhere else is bone, ink and hairlines.

Zips jam.

INTER TIGHT · BODY

PLATE IV · 100 MM SEAM

JETBRAINS MONO · PLATE FURNITURE

INSTRUMENT SERIF · DISPLAY

Everywhere a zip is
currently *tolerated*.

Tape width	9 mm
Barb pitch	1.6 mm
Material	Single polymer
Shear hold	42 N / 100 mm
Peel release	3.1 N
Closing force	1.8 N, any point
Cycles	20,000 to 90%
Metal content	None
Operation	One hand
Acoustic	Effectively silent

ALL VALUES ARE DESIGN TARGETS,
NOT MEASURED RESULTS.

Adaptive clothing

One hand, no pinch grip, no help from anyone else.
The case the zip has never served.

Outerwear

Full-length front closure with no start point. Gloves
on, in the dark, at speed.

Luggage

Nothing to burst under load. Shear resistance rises
with the weight inside the bag.

Medical and outdoor

No metal, nothing to harbour contamination, no slider
to freeze or lose.

Built whole. *Labelled honestly.*

How it is built

Hand-drawn, not libaried

Every mechanism is SVG driven by `requestAnimationFrame`, with no animation library touching them. Each rig idles when settled and when offscreen, so a page carrying five live mechanisms costs nothing while you read it. 63 kB of gzipped JavaScript, 4 kB of CSS.

What is real

The biology and the material case

A feather vane genuinely is held by hooked barbules meeting ridged ones, and a bird does re-link them in one pass. Mono-material design is already how the recycling side of the industry thinks about closures.

What is not

Nothing has been tooled

VANE has not been moulded or pull tested. Every number here is a design target we set ourselves, not a result off a rig. The site says so in its own colophon rather than burying it, and so does this document.

Made by ac.

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